1. Clinical Question & Core Hypothesis

Clinical Question

Why do tumors with high stromal proportion (high TSR) consistently show poorer outcomes in MIBC — is the stroma simply a passive correlate or does it actively create a microenvironment that impairs anti■tumor control? 2) Tier 1 path used: single provided path (TSR → stroma proportion → prognosis); I built directly on this path because it gives the strongest mechanistic backbone linking a morphologic spatial measure to outcome. 3) Tier 2 evidence: while no explicit Tier 2 gene/path data were provided, published mechanistic work (CAF activity, collagen/ECM remodeling, and immune exclusion) supports the claim that stroma can be biologically active and immune■suppressive; these concepts motivate the mechanistic framing rather than additional gene anchors. 4) Endpoint choice: Overall survival (OS) was chosen because it is the most robust, clinically definitive endpoint available for prognostic biomarker studies and aligns with prior TSR literature in solid tumors. 5) Deliberate exclusions: I excluded molecular/gene anchors because none were provided by the Ontologist (to avoid inventing genes); I also excluded multi■feature models and therapy■response endpoints to keep the hypothesis sharply focused on mechanism linking morphology to survival?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), a high stromal-to-tumor ratio (TSR) measured at the invasive tumor front reflects a cancer-associated fibroblast (CAF)– and extracellular matrix (ECM)–dense microenvironment that actively promotes immune exclusion and tumor invasion, thereby accelerating progression. This stromal biology leads to poorer overall survival (OS) independent of standard clinicopathologic factors (stage, grade, age, and treatment). Furthermore, TSR identifies a biologically distinct, stroma-driven high-risk subgroup of MIBC patients who may benefit from stroma-targeted or combination therapies that aim to normalize the ECM and restore anti-tumor immunity. The hypothesis will be tested by: (i) demonstrating that higher TSR correlates with reduced CD8+ T cell infiltration and with elevated CAF/ECM markers at the invasive front; (ii) showing that TSR remains independently prognostic for OS after adjusting for confounders; and (iii) validating the finding in an independent cohort and exploring mediation by immune exclusion rather than direct tumor intrinsic factors.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Primary endpoint	OS

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions